

Section 3.5

I. Chain Rule – Multiplier Effect

II. Examples of Compositions

III. Forms of the Chain Rule

a. $y = f(u), u = g(x) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$

b. $y = f(u(x)) \Rightarrow \frac{d f(u(x))}{dx} = \frac{d f(u)}{du} \frac{du}{dx}$

c. $y = f(u(x)) \Rightarrow (f(u(x)))' = f'(u(x))u'(x)$

Examples

IV. Extended Derivative Formulas

a. Power Rule $\frac{d}{dx} u^n = n u^{n-1} \frac{du}{dx}$

b. Trig Rules $\frac{d}{dx} \sin u = \cos u \frac{du}{dx}$
 $\frac{d}{dx} \cos u = -\sin u \frac{du}{dx}$

c. Exp/Log Rules $\frac{d}{dx} e^u = e^u \frac{du}{dx}$
 $\frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx}$